UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY

DEPARTMENT OF COMPUTING AND TECHNOLOGY

EASTER 2025 SEMESTER **TEST PRACTICAL**

PROGRAM: *BSIT*

YEAR: 1 SEMESTER: 2

COURSE CODE: *ICT2106*

COURSE NAME: *LOCAL AREA COMPUTER NETWORKING*

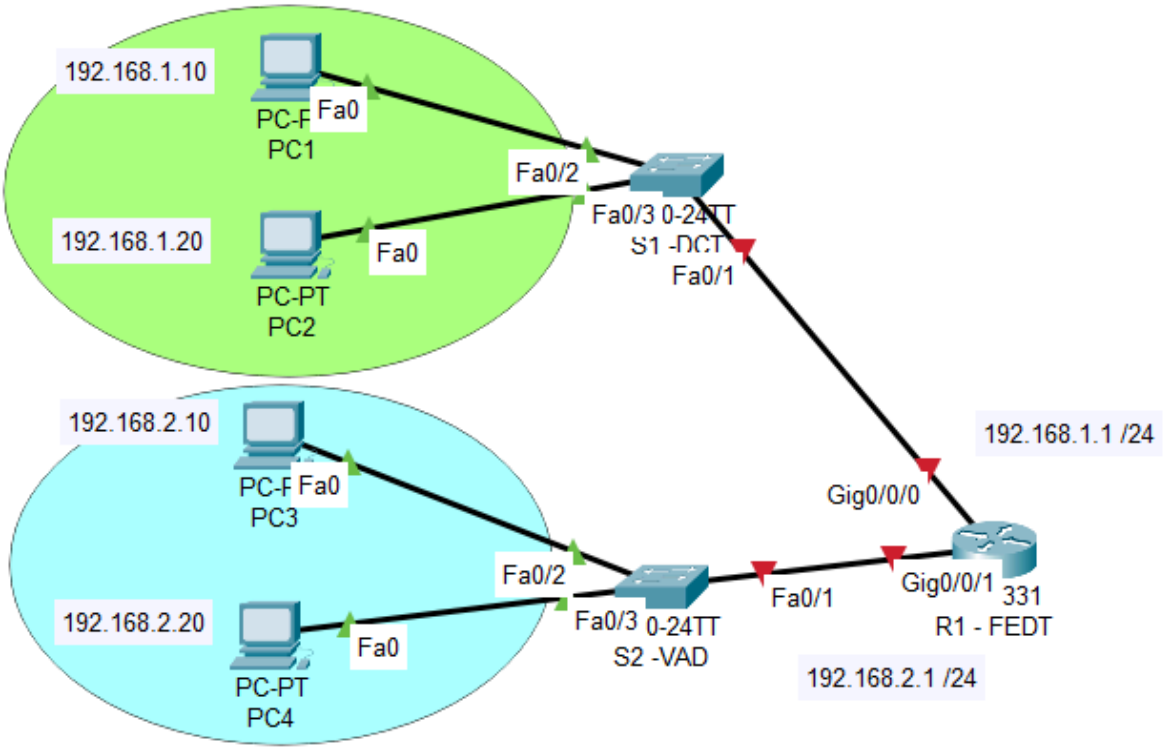
TEST DATE: 11th March 2025

TIME ALLOWED: *2 Hours*

Test Instructions

a) Attempt all questions

You are a network technician tasked with setting up a **small enterprise network.** The network consists of:



- **1 Router (R1 - FEDT)**
- **2 Switches (S1 - DCT, S2 - VAD)**
- **4 PCs (PC1, PC2, PC3, PC4)**

Your goal is to configure the network for **basic connectivity, security, and troubleshooting**.

TASKS

1. Physical and Logical Connections (10 Marks)

2. Device Naming & Security (10 Marks)

Set the **following hostnames** on devices:

- **Router R1 → FEDT**
- **Switch S1 → DCT**
- **Switch S2 → VAD**

Set **passwords for security**:

- **User Config Mode (Line Console 0) → cisco**
- **Virtual Terminal Lines (VTY) → class**
- **Privileged EXEC Mode (Enable Password) → cisco**

Configure a **MOTD (Message of the Day) banner** with your name.

Encrypt all passwords using the encryption commands

Save the configuration

3. Router Configuration (10 Marks)

Configure **Router R1 (FEDT)** with the following:

- **G0/0 → 192.168.1.1 /24**
- **G0/1 → 192.168.2.1 /24**
- **Activate the interfaces**
- **Verify the status**

Save the configuration.

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4. PC Configuration & Connectivity (10 Marks)

Assign **static IP addresses** as per the table:

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.1.10	255.255.255.0	192.168.1.1
PC2	192.168.1.20	255.255.255.0	192.168.1.1
PC3	192.168.2.10	255.255.255.0	192.168.2.1
PC4	192.168.2.20	255.255.255.0	192.168.2.1

Verify configurations

5. Testing and Troubleshooting (5 Marks)

- Verify network connectivity
- Identify and **fix errors** (if any).

6. Bonus: Advanced Security (5 Marks)

- Enable **SSH** access on Router R1
- Save configurations.

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